

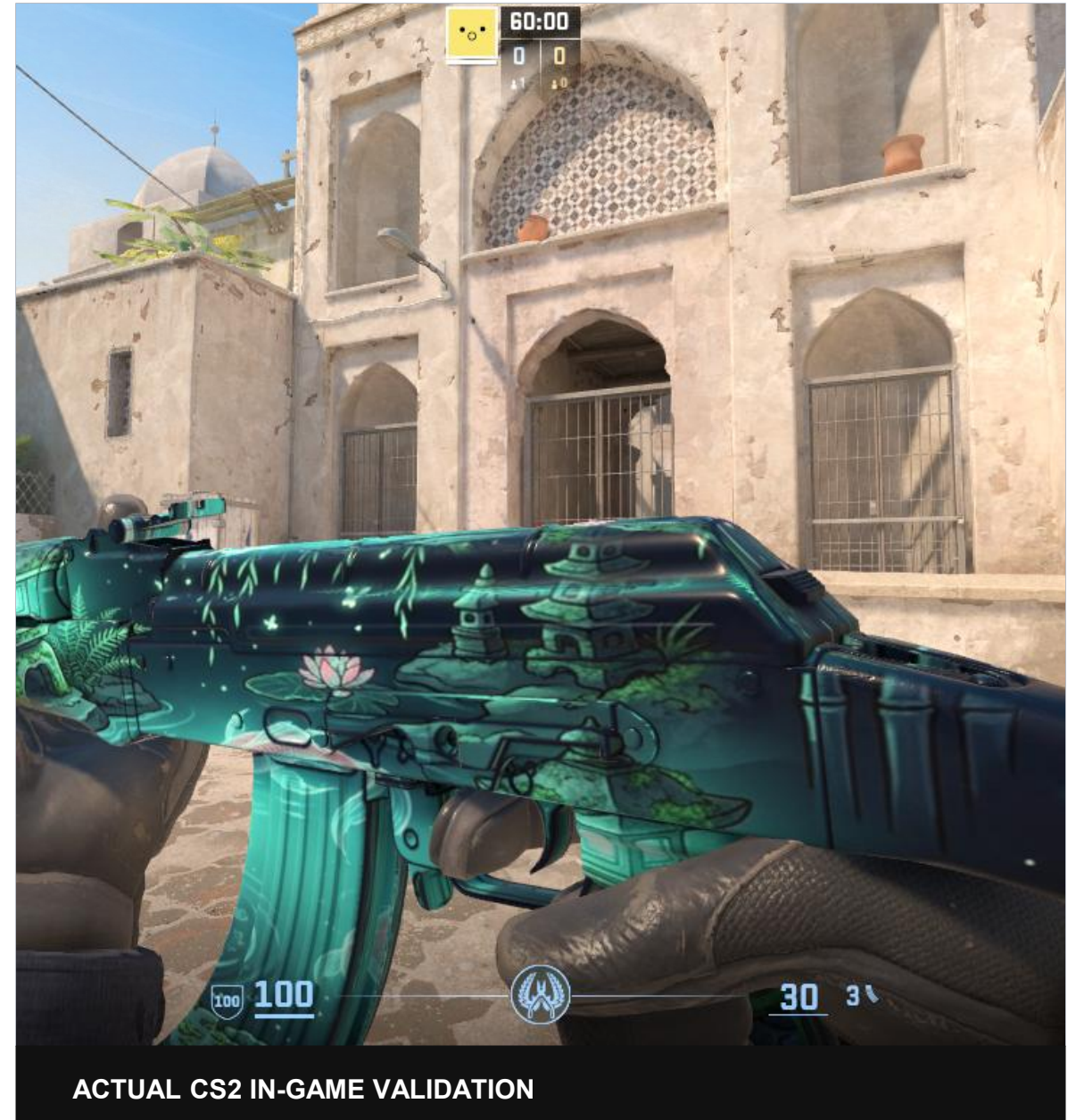
SkinSmith

A Constraint-Aware Generative Agent for Game Weapon Skin Design

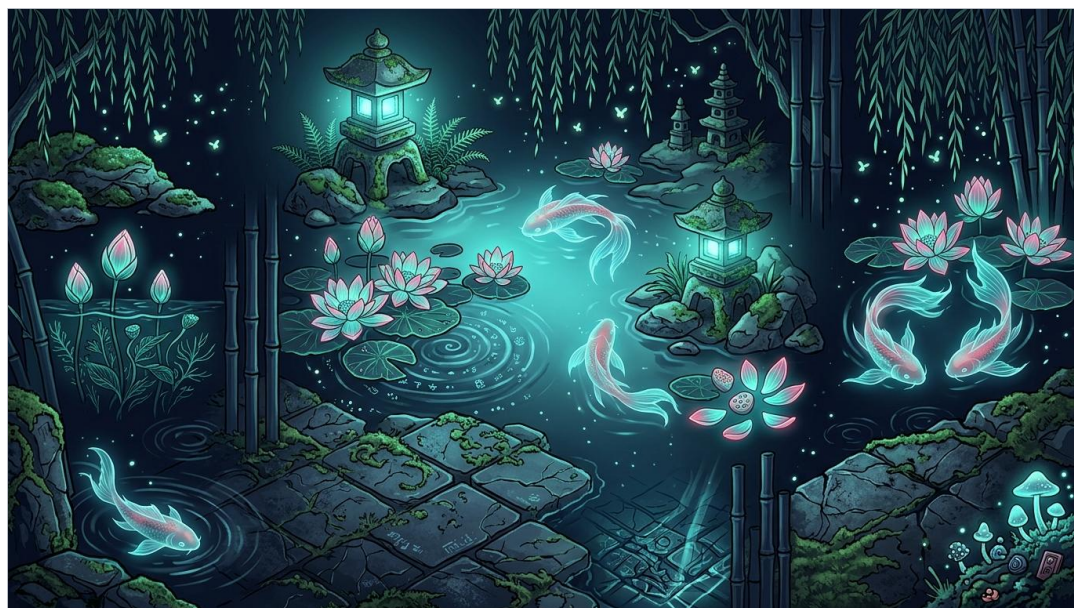
A Counter-Strike 2 case study in human-in-the-loop mapped-asset evaluation

YUAN Ye · CHEN Yuhong · Ben Tangjie
Wang Zhengye · WANG Lihui

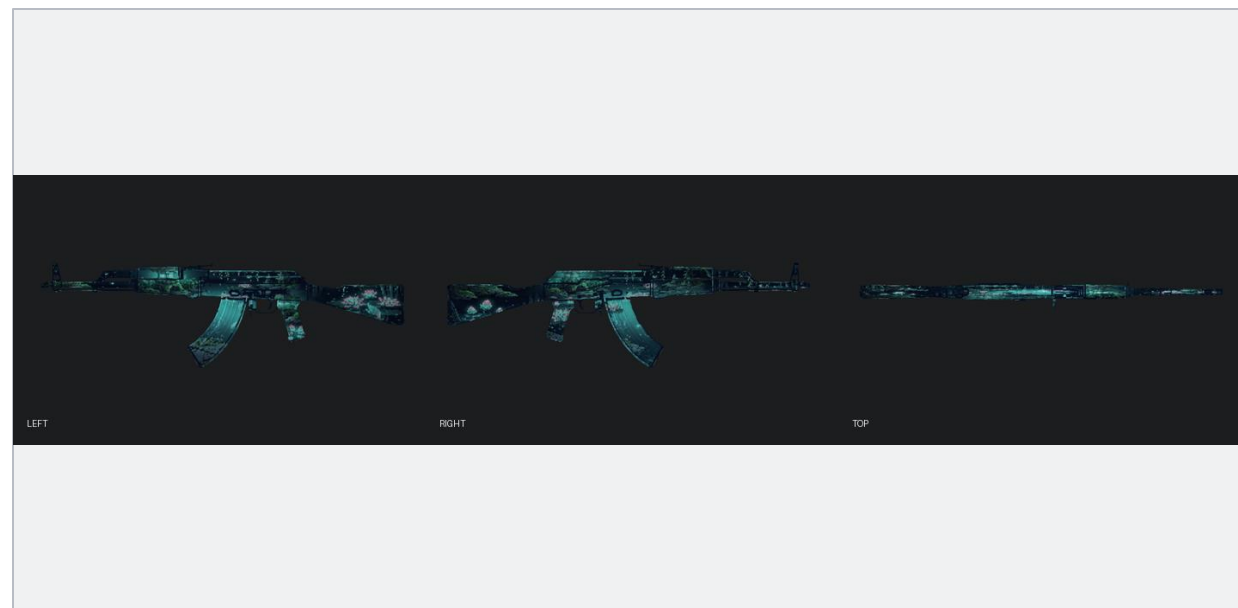
Supervisor: Prof. Yu Yang



Source-image quality is not deployment evidence



SOURCE IMAGE



RECONSTRUCTED ASSET VIEWS

UV islands split and rotate the visible surface. SkinSmith therefore judges the source and fixed left/right/top reconstructions together.

Three checkpoints preserve user authority

01 · THEME WORLD

Confirm what “garden” means

Agent expands subjects, symbols, palette, atmosphere and exclusions; the user confirms the theme world.

02 · ART DIRECTION

Choose a distinct visual route

User selects one of four materially different text directions; model ranking remains advisory.

03 · MAPPED ARTWORK

Judge source and views together

Four source + mapped-view cards remain visible so technical scoring cannot replace aesthetic choice.

Formal execution begins only after the user resolves all three checkpoints.

From “garden” to a locked visual direction



1 ONE-WORD INPUT

garden — a transferable concept, not a weapon-specific prompt.

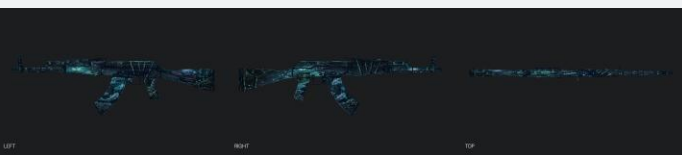
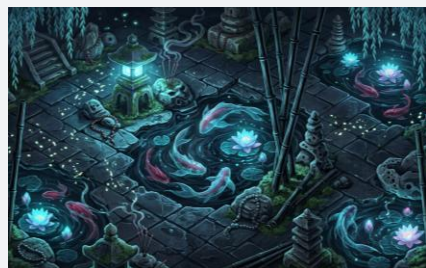
2 CHECKPOINT 1

Midnight Serenity: lotus, koi, pagoda, bamboo and cyan-jade light.

3 CHECKPOINT 2

Spectral Movement selected from four distinct visual routes.

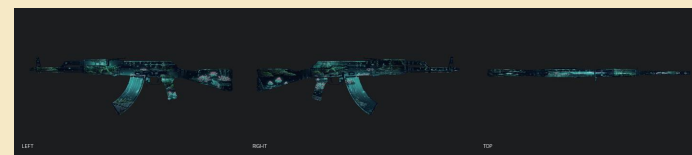
Technical rank is not aesthetic choice



01 · Narrative Panorama

technical total 0.8049

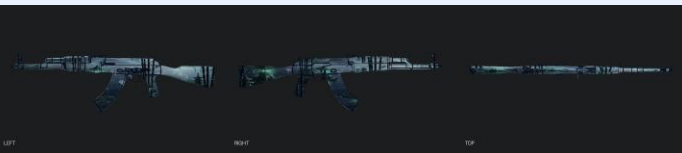
FEASIBLE



02 · Ornamental Tapestry

technical total 0.8293

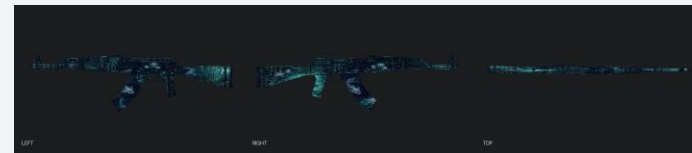
HUMAN SELECTED



03 · Atmospheric World

technical total 0.8955

AUTO TOP SCORE



04 · Kinetic Myth

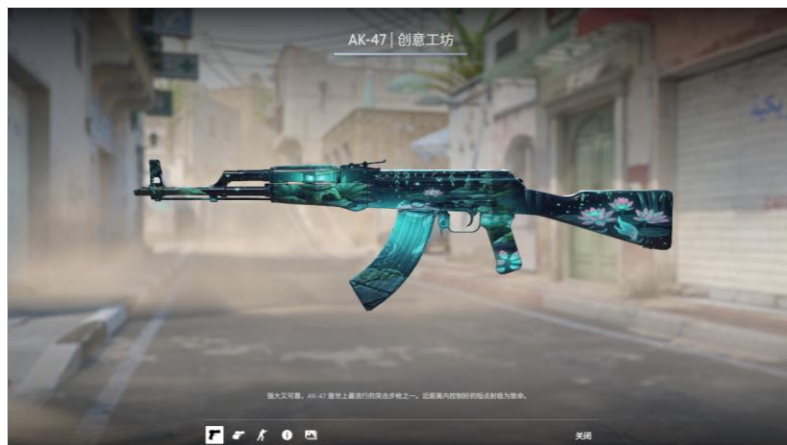
technical total 0.7730

FEASIBLE

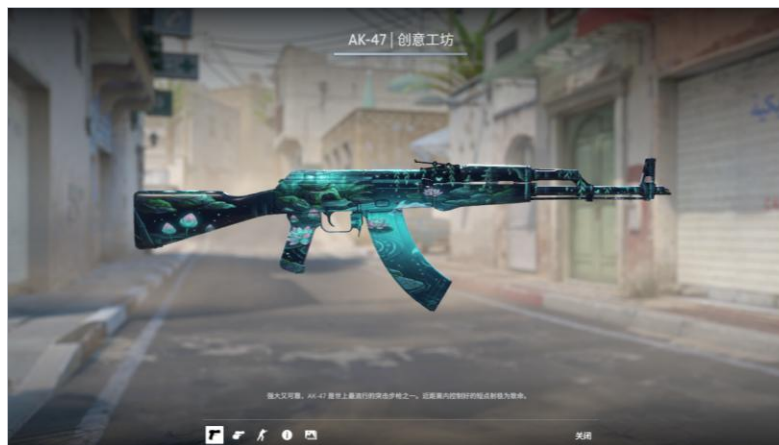
Metrics screen feasibility; people choose the design.

artwork_03 ranked highest technically; the user preferred artwork_02.

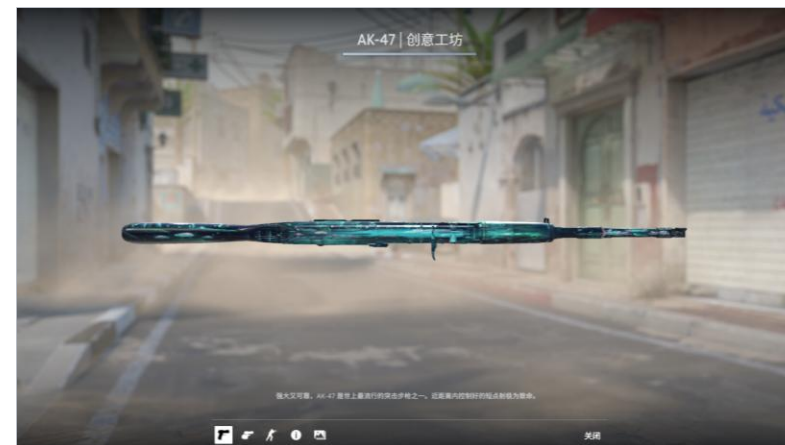
Fixed views validate the selected color texture



LEFT



RIGHT



TOP

0.001246

Formal asset-seam error

0.811677

Formal multi-view score

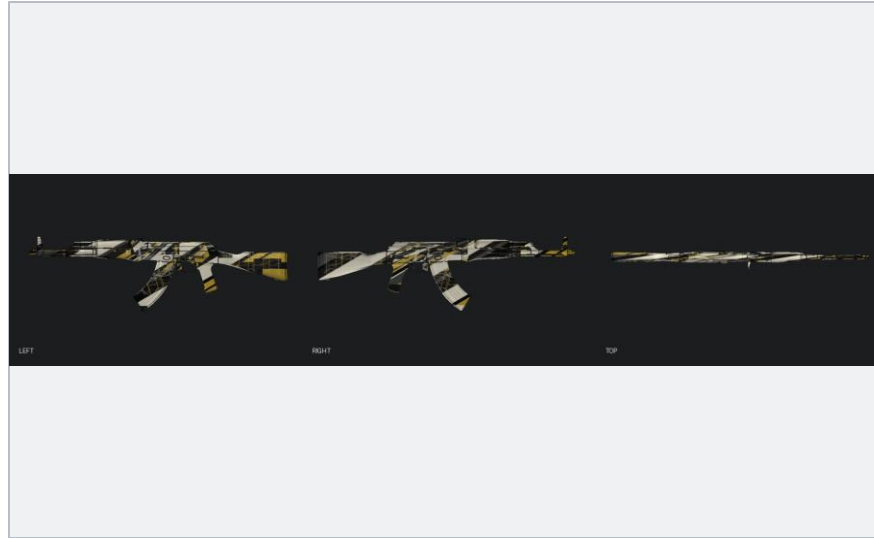
PASS

Workshop left/right/top review

Route B improves both measures in all four pairs



ROUTE A · PATTERN-FIRST BASELINE



ROUTE B · WEAPON-SPACE

99.77%

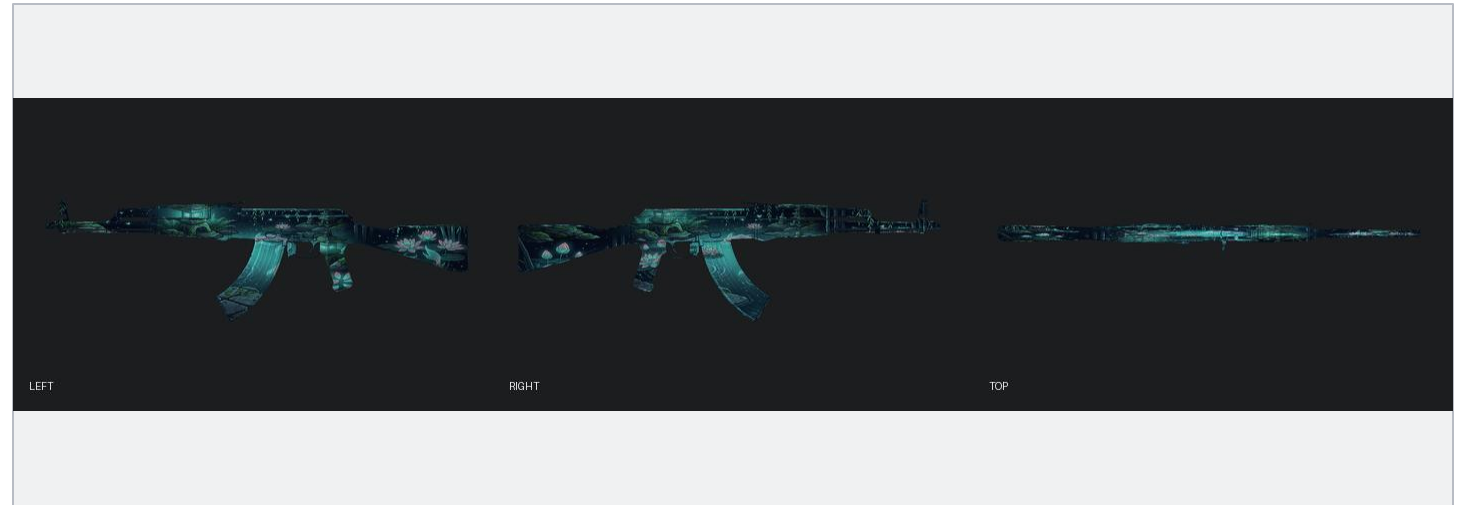
Mean seam reduction
0.26997 → 0.00063

+0.060

Mean mapped-view
score
0.76898 → 0.82911

The result is about reconstruction quality—not whether the flat UV atlas looks attractive.

Human choice reaches the engine



garden → Spectral Movement → artwork_02

0.001246

Formal asset-seam error

0.811677

Formal multi-view score

0.817374

Final total score

**ENGINE
PASS**

CS2 verified

Hard gates and rollback prevent invalid choices

01

GATE

Reject candidates that violate the asset-seam threshold.

02

SELECT

Rank only feasible candidates by mapped quality and preference.

03

REFINE ONCE

Apply one local correction when the diagnosis is justified.

04

ROLL BACK

Retain Route B when correction fails the improvement rule.

Route C score change

-0.016233

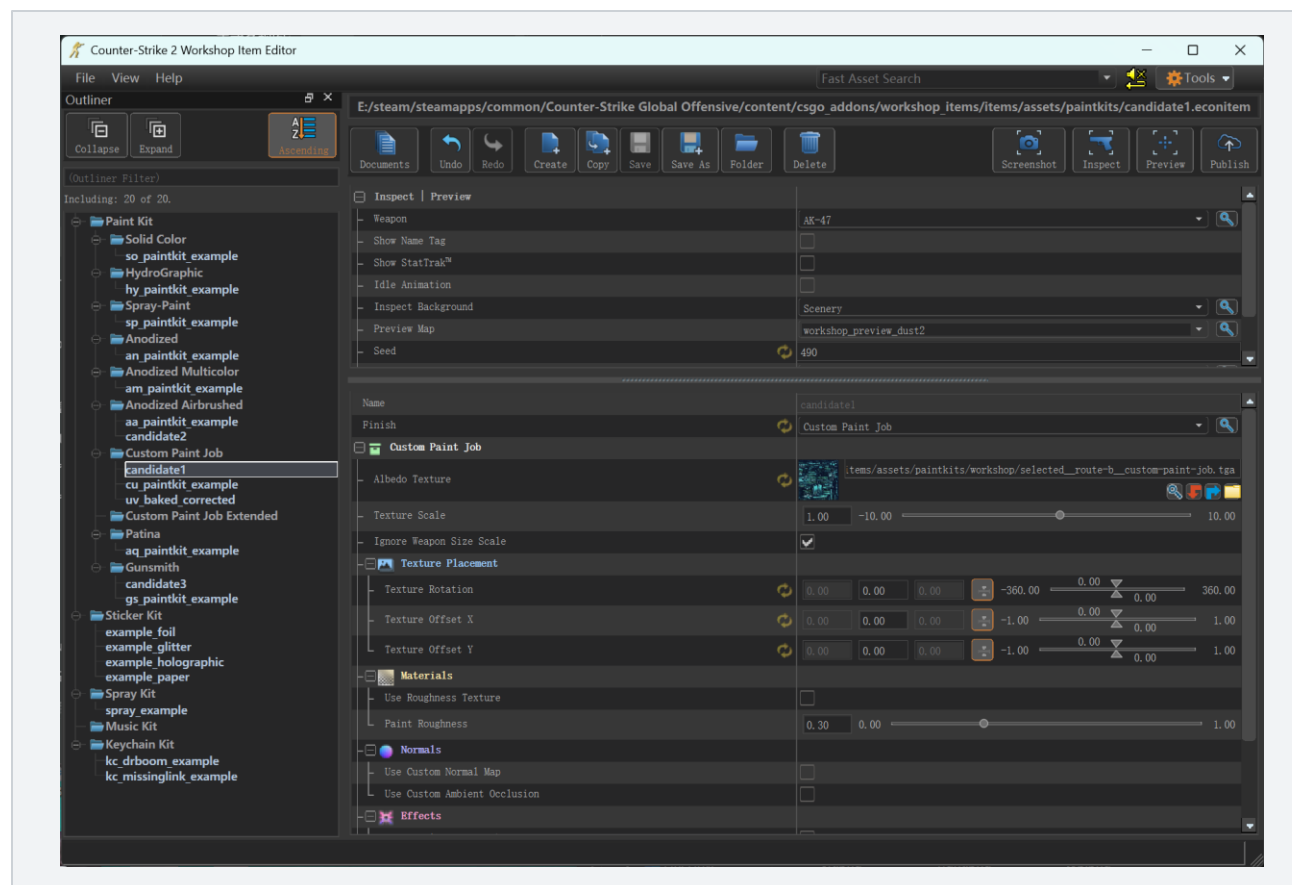
Correction rejected · Route B retained

Workshop settings preserve the export contract

FORMAL EXPORT

2048
24-bit RGB
TGA

Exact selected source
SHA-256 preserved
PNG/TGA pixels identical

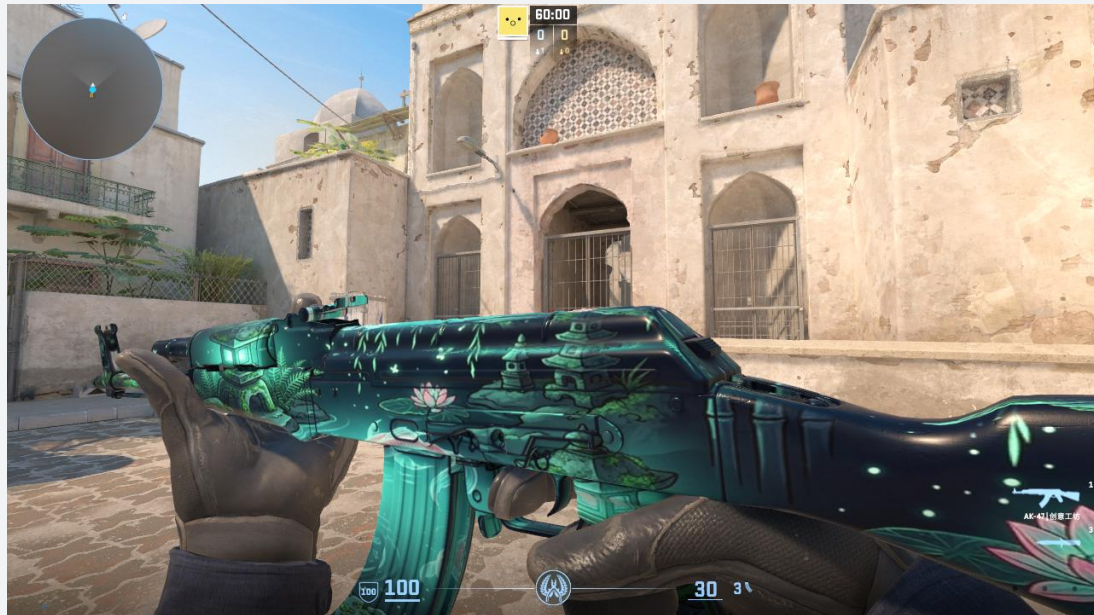


CUSTOM PAINT JOB

Scale 1.0
Rotation 0°
Offset X 0
Offset Y 0
Ignore weapon size
scale: ON
Custom normal
map: OFF

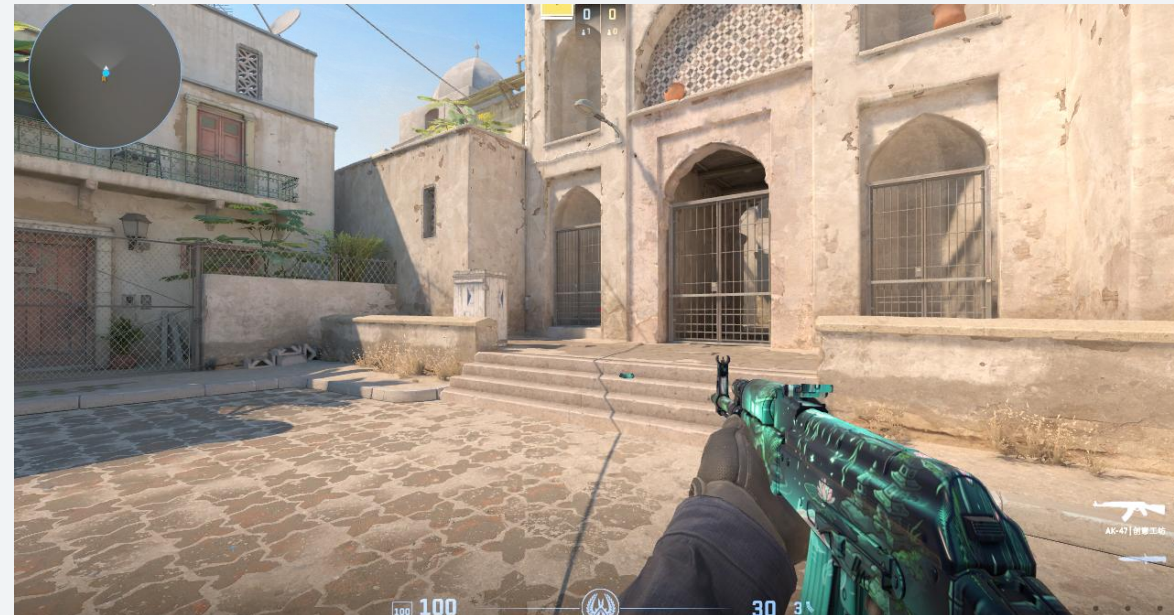
Base color only

In-game proof is color placement—not full PBR



DETAIL INSPECTION

Motif placement, coverage and readability remain visible at close range.



PLAY VIEW

The texture remains legible in a normal held-weapon view.

Not claimed: Agent-generated normal, roughness, metallic, height or displacement channels.

The Agent is complete for the defined AK-47 case

3

Human checkpoints separate meaning, direction and mapped aesthetic choice

4 candidates

All remain visible; technical rank does not replace user preference

1

Bounded correction produced one explicit, evidence-backed rollback

Main conclusion

Technical scoring guides feasibility; human checkpoints retain aesthetic authority. Acceptance belongs to the reconstructed asset and its preserved evidence.

NEXT RESEARCH LAYER

PBR channels
Engine capture
More assets and users

Questions

Five connected work packages

YUAN Ye — project lead, architecture and integration

CHEN Yuhong — creative planning and literature

Ben Tangjie — generation backends and reproducibility

Wang Zhengye — asset, UV and rendering validation

WANG Lihui — evaluation, reporting and delivery QA

github.com/DID-FIELD/SDSC6002-SkinSmith

